

## 12.3 Prediction intervals

Often the quantity of interest is an observation not yet made: tomorrow's stock loss, or the reading of the next specimen. An interval for it must allow for the error in the estimated mean, which shrinks as data accumulate, and the error of the new observation itself, which does not. The first depends on where the new point lies relative to the data; the second makes prediction intervals depend on normality in a way that confidence intervals do not.

### 12.3.1 Predicting a new observation

Let the data follow the normal linear model of [Section 12.1](#), and let

$$Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0, \quad \varepsilon_0 \sim N(0, \sigma^2), \quad \varepsilon_0 \text{ independent of } \mathbf{Y}, \quad (12.3.1)$$

be a future observation at a regressor vector  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ . The requirement  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$  makes the mean  $\mathbf{x}_0^\top \boldsymbol{\beta}$  estimable. Without it, Proposition 12.1.3 shows that nothing useful can be said. We call

$$h_0 = \mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0$$

the **leverage** of the new point. When  $\mathbf{x}_0$  is the  $i$ th row of  $\mathbf{X}$ ,  $h_0$  is the diagonal entry  $h_{ii}$  of  $\mathbf{M}$  (Proposition 6.8.1), and  $\sigma^2 h_0$  is always the variance of  $\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$  (Theorem 7.5.1(a)).

#### Theorem 12.3.1. Prediction interval

Under [Eq. 12.3.1](#), put  $\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ .

- (a)  $Y_0 - \hat{Y}_0 \sim N(0, \sigma^2(1 + h_0))$ , it is independent of  $s^2$ , and

$$\frac{Y_0 - \hat{Y}_0}{s\sqrt{1 + h_0}} \sim t(n - r).$$

- (b) The interval  $\hat{Y}_0 \pm t_{n-r, \alpha/2} s\sqrt{1 + h_0}$  contains  $Y_0$  with probability exactly  $1 - \alpha$ , the probability being computed over  $\mathbf{Y}$  and  $Y_0$  jointly, at every  $(\boldsymbol{\beta}, \sigma^2)$ .
- (c) More generally, if  $\bar{Y}_0$  is the mean of  $m$  independent new observations at  $\mathbf{x}_0$ , each distributed as in [Eq. 12.3.1](#), then  $\hat{Y}_0 \pm t_{n-r, \alpha/2} s\sqrt{1/m + h_0}$  contains  $\bar{Y}_0$  with probability exactly  $1 - \alpha$ . As  $m \rightarrow \infty$  this interval becomes the confidence interval  $\hat{Y}_0 \pm t_{n-r, \alpha/2} s\sqrt{h_0}$  for  $\mathbf{x}_0^\top \boldsymbol{\beta}$ .

*Proof.* We prove (c), of which (a) and (b) are the case  $m = 1$ . The new observations are independent of  $\mathbf{Y}$ , and  $\bar{Y}_0 \sim N(\mathbf{x}_0^\top \boldsymbol{\beta}, \sigma^2/m)$ . The vector  $(\bar{Y}_0, \mathbf{Y}^\top)^\top$  is normal with independent blocks, so  $\bar{Y}_0 - \hat{Y}_0$ , a linear function of it, is normal (Theorem 3.3.1) with mean  $\mathbf{x}_0^\top \boldsymbol{\beta} - \mathbf{x}_0^\top \boldsymbol{\beta} = 0$  and variance  $\sigma^2/m + \sigma^2 h_0$ , the two terms adding because  $\bar{Y}_0$  and  $\hat{Y}_0$  are independent. For independence from  $s^2$ :  $\hat{Y}_0$  is a function of  $\mathbf{M}\mathbf{Y}$ , and  $\mathbf{M}\mathbf{Y}$  and SSE are independent (Theorem 7.5.1(c)). Since  $\bar{Y}_0$  is independent of  $\mathbf{Y}$ , for any events,

$$\begin{aligned} \Pr(\bar{Y}_0 \in A, \mathbf{M}\mathbf{Y} \in B, \text{SSE} \in D) \\ = \Pr(\bar{Y}_0 \in A) \Pr(\mathbf{M}\mathbf{Y} \in B) \Pr(\text{SSE} \in D), \end{aligned}$$

so  $(\bar{Y}_0, \mathbf{M}\mathbf{Y})$  is independent of SSE, and so is  $\bar{Y}_0 - \hat{Y}_0$ . Dividing the standardized normal

variable by  $\sqrt{(\text{SSE}/\sigma^2)/(n-r)}$  gives  $t(n-r)$  (Definition 4.1.3), and the interval statement is  $\Pr\{|T| \leq t_{n-r,\alpha/2}\} = 1 - \alpha$ . The limit as  $m \rightarrow \infty$  is immediate, and the limiting interval is Eq. 12.1.1 with  $\lambda = \mathbf{x}_0$ . ■

Part (a) was sketched in Exercise (B2). A *prediction* interval is for a random variable, and its probability refers to the joint experiment: collect the data, compute the interval, observe  $Y_0$ . Given the data, the probability that the interval contains  $Y_0$  is random,

$$\Pr(Y_0 \in \text{interval} \mid \mathbf{Y}) = \Phi\left(\frac{\mathbf{x}_0^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) + t s \sqrt{1 + h_0}}{\sigma}\right) - \Phi\left(\frac{\mathbf{x}_0^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - t s \sqrt{1 + h_0}}{\sigma}\right),$$

with  $t = t_{n-r,\alpha/2}$  and  $\Phi$  the standard normal distribution function, and it is only its expectation that equals  $1 - \alpha$ .

In the variance  $\sigma^2(1 + h_0)$ , the leverage  $h_0$  is the price of not knowing  $\boldsymbol{\beta}$ , of order  $r/n$  at a typical point (Proposition 6.8.1). The 1 is the new error, which no amount of data removes: the prediction interval never becomes shorter than about  $2z_{\alpha/2}\sigma$ , while the interval for the mean shrinks to a point.

#### Example. Predicting a state's murder rate

In the murder-rate regression, consider a state at the centroid of the regressors. Its leverage is  $h_0 = 1/n = 0.0200$  (see Proposition 12.3.2), the fitted rate is 4.514, and the 95% intervals are

mean response: (4.090, 4.938),      new state: (1.487, 7.541).

The prediction interval is 7.14 times as long as the interval for the mean, and nearly all of its length, 6.053 per 100,000, comes from  $s$ .

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statcrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s = np.sqrt(np.sum((y - X @ beta_hat) ** 2) / (n - p))
q = stats.t.ppf(0.975, n - p)
h = np.einsum("ij,jk,ik->i", X, C, X) # leverages of the 50 states

def intervals(x0):
    """Leverage h0, 95% interval for the mean response, 95% prediction interval."""
    h0 = x0 @ C @ x0
    fit = x0 @ beta_hat
    ci = (fit - q * s * np.sqrt(h0), fit + q * s * np.sqrt(h0))
    pi = (fit - q * s * np.sqrt(1 + h0), fit + q * s * np.sqrt(1 + h0))
```

```

return h0, fit, ci, pi

x_centre = X.mean(axis=0)                # the average state
x_hidden = np.array([1.0, 20.0, 19.0, 60.0]) # each coordinate inside its
observed range
for name, x0 in [("centre", x_centre), ("hidden", x_hidden)]:
    h0, fit, ci, pi = intervals(x0)
    print(f"{name}: h0 = {h0:.4f}, fit {fit:.3f}, mean ({ci[0]:.3f}, {ci[1]:.3f}), "
          f"new state ({pi[0]:.3f}, {pi[1]:.3f})")
print(f"largest leverage in the data: {h.max():.4f}")

```

[Run this code online](#)

## 12.3.2 Leverage and extrapolation

### Proposition 12.3.2. Leverage of a new point

Let  $\mathbf{x}_0 = \mathbf{X}^\top \boldsymbol{\rho} \in \mathcal{C}(\mathbf{X}^\top)$  and  $h_0 = \mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0$ .

- (a)  $h_0 = \min\{\|\mathbf{a}\|^2 : \mathbf{X}^\top \mathbf{a} = \mathbf{x}_0\}$ , and the minimum is attained only at  $\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ .
- (b) If  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$  has full column rank,  $\mathbf{x}_0 = (1, \mathbf{x}_{01}^\top)^\top$ ,  $\bar{\mathbf{x}}$  is the vector of column means of  $\mathbf{X}_1$  and  $\tilde{\mathbf{X}}_1 = \mathbf{X}_1 - \mathbf{1}\bar{\mathbf{x}}^\top$ , then

$$h_0 = \frac{1}{n} + (\mathbf{x}_{01} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{01} - \bar{\mathbf{x}}).$$

- (c) If further columns are added,  $\mathbf{X}_+ = [\mathbf{X}, \mathbf{Z}]$ , and  $\mathbf{x}_{0+} = (\mathbf{x}_0^\top, \mathbf{z}_0^\top)^\top \in \mathcal{C}(\mathbf{X}_+^\top)$ , then the leverage of  $\mathbf{x}_{0+}$  in the enlarged model is at least  $h_0$ .

*Proof.*

- (a) Let  $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$ . Then  $\mathbf{X}^\top \mathbf{M}\mathbf{a} = \mathbf{X}^\top \mathbf{a} = \mathbf{x}_0 = \mathbf{X}^\top \mathbf{M}\boldsymbol{\rho}$ , so  $\mathbf{M}\mathbf{a} - \mathbf{M}\boldsymbol{\rho}$  lies in  $\mathcal{C}(\mathbf{X})$  and is orthogonal to  $\mathcal{C}(\mathbf{X})$ ; hence  $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ . By Pythagoras,  $\|\mathbf{a}\|^2 = \|\mathbf{M}\boldsymbol{\rho}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2 \geq \|\mathbf{M}\boldsymbol{\rho}\|^2$ , with equality iff  $\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ , and  $\|\mathbf{M}\boldsymbol{\rho}\|^2 = \boldsymbol{\rho}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \boldsymbol{\rho} = h_0$  by Theorem 6.3.6.
- (b)  $\mathbf{X} = [\mathbf{1}, \tilde{\mathbf{X}}_1] \mathbf{A}$  with  $\mathbf{A} = \begin{pmatrix} 1 & \bar{\mathbf{x}}^\top \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$ , which is invertible. Since  $\mathbf{1}^\top \tilde{\mathbf{X}}_1 = \mathbf{0}^\top$ ,  $\mathbf{X}^\top \mathbf{X} = \mathbf{A}^\top \text{diag}(n, \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1) \mathbf{A}$ , and so  $h_0 = \mathbf{w}^\top \text{diag}(1/n, (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}) \mathbf{w}$  with  $\mathbf{w} = \mathbf{A}^{-\top} \mathbf{x}_0$ . Solving  $\mathbf{A}^\top \mathbf{w} = \mathbf{x}_0$  gives  $\mathbf{w} = (1, (\mathbf{x}_{01} - \bar{\mathbf{x}})^\top)^\top$ .
- (c) Every  $\mathbf{a}$  with  $\mathbf{X}_+^\top \mathbf{a} = \mathbf{x}_{0+}$  also satisfies  $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$ . A minimum over a smaller set is at least as large, so (a), applied to both models, gives the claim. ■

Part (a) is the Gauss–Markov theorem in disguise. Part (b) makes leverage a distance: the quadratic form is  $(n - 1)^{-1}$  times the squared Mahalanobis distance of  $\mathbf{x}_{01}$  from the centroid of the regressors, as for the observed points (Proposition 6.8.2). So the interval for the mean is shortest at the centroid, where  $h_0 = 1/n$ , and widens along ellipsoidal contours. Part (c) says a larger model never predicts more precisely at a given point, though it may predict with less bias; Chapter 29 weighs the two.

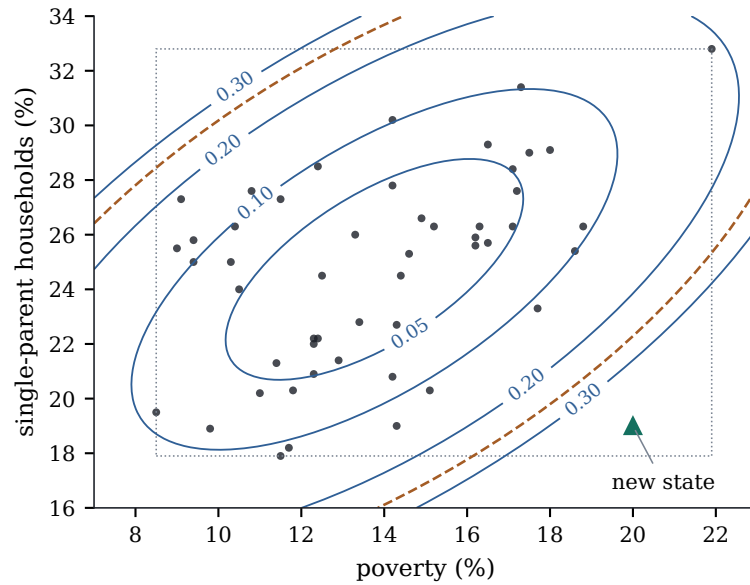


Figure 12.3.1: Contours of the leverage  $h_0$  of a new state (urbanization at its mean), with the 50 states, the box of observed ranges (dotted) and the largest observed leverage (dashed). The marked state is inside the box but far from the data.

With several correlated regressors, a point can lie inside the range of every regressor and still be far from the data, because it breaks their correlation. This **hidden extrapolation** is detected by the leverage, not by the separate ranges.

#### Example. Hidden extrapolation

Poverty ranges from 8.5% to 21.9% over the 50 states, and single-parent households from 17.9% to 32.8%. A state with poverty 20%, single-parent households 19% and 60% urban is inside every range, but poor with few single parents, a combination absent from the data. Its leverage is  $h_0 = 0.435$ , far above the largest leverage among the actual states, 0.260 (Mississippi). Figure 12.3.1 shows the contours of  $h_0$  over the two regressors, with urbanization held at its mean.

The fitted rate is 3.817. The 95% interval for the mean is (1.840, 5.794), longer than at the centroid by the factor  $\sqrt{0.435/0.0200} = 4.66$ . The prediction interval, (0.227, 7.407), is only 1.186 times as long as at the centroid, because it is dominated by  $s$ . The prediction interval barely registers the extrapolation, and the real danger, that the model may be wrong where there are no data to check it, is in neither formula. If urbanization is dropped from the model the leverage of this point falls to 0.287, as Proposition 12.3.2(c) requires.

### 12.3.3 When the errors are not normal

The estimate  $\hat{Y}_0$  is a weighted sum of  $n$  errors, approximately normal by a central limit theorem when no weight dominates (Theorem 19.3.2). The new error  $\varepsilon_0$  is a single draw, and its distribution enters the prediction interval unchanged, however large  $n$  is.

#### Proposition 12.3.3. Prediction intervals without normality

Consider a sequence of full-rank models  $\mathbf{Y}_n = \mathbf{X}_n\boldsymbol{\beta} + \boldsymbol{\varepsilon}_n$  with  $p$  fixed and  $n \rightarrow \infty$ , and

a fixed  $\mathbf{x}_0$  whose leverage  $h_{0,n} = \mathbf{x}_0^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{x}_0$  tends to zero. Let  $\varepsilon_1, \dots, \varepsilon_n$  and  $\varepsilon_0$  be independent and identically distributed with mean 0, variance  $\sigma^2$ , finite fourth moment and a continuous distribution function, and let  $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ . Then the coverage of the nominal  $1 - \alpha$  prediction interval  $\hat{Y}_0 \pm t_{n-p, \alpha/2} s \sqrt{1 + h_{0,n}}$  satisfies

$$\begin{aligned} \Pr\{|Y_0 - \hat{Y}_0| \leq t_{n-p, \alpha/2} s \sqrt{1 + h_{0,n}}\} \\ \longrightarrow \Pr\{|\varepsilon_0| \leq z_{\alpha/2} \sigma\}, \end{aligned}$$

where  $z_{\alpha/2}$  is the upper  $\alpha/2$  point of  $N(0, 1)$ . Likewise the probabilities of missing above and below tend to  $\Pr\{\varepsilon_0 > z_{\alpha/2} \sigma\}$  and  $\Pr\{\varepsilon_0 < -z_{\alpha/2} \sigma\}$ .

*Proof.* Write  $T_n = (Y_0 - \hat{Y}_0)/(s \sqrt{1 + h_{0,n}})$ ,  $t_n = t_{n-p, \alpha/2}$  and  $E = \varepsilon_0/\sigma$ . First,  $Y_0 - \hat{Y}_0 = \varepsilon_0 - \mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ , and  $\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$  has mean zero and variance  $\sigma^2 h_{0,n} \rightarrow 0$  (Theorem 5.5.1), so it tends to zero in probability by Chebyshev's inequality. Second,  $s \rightarrow \sigma$  in probability (Corollary 5.6.4), and  $t_n \rightarrow z_{\alpha/2}$ . Hence  $D_n = T_n - E \rightarrow 0$  in probability. Fix  $\eta > 0$ . On the event  $\{|D_n| \leq \eta/2, |t_n - z_{\alpha/2}| \leq \eta/2\}$ , whose probability tends to one,

$$|E| \leq z_{\alpha/2} - \eta \implies |T_n| \leq t_n \implies |E| \leq z_{\alpha/2} + \eta.$$

So  $\Pr\{|E| \leq z_{\alpha/2} - \eta\} \leq \liminf \Pr\{|T_n| \leq t_n\} \leq \limsup \Pr\{|T_n| \leq t_n\} \leq \Pr\{|E| \leq z_{\alpha/2} + \eta\}$ . The distribution function of  $E$  is continuous, so letting  $\eta \rightarrow 0$  gives the limit. The one-sided statements are proved the same way. ■

The limit is the nominal  $1 - \alpha$  only if  $\Pr\{|\varepsilon_0| \leq z_{\alpha/2} \sigma\} = 1 - \alpha$ , a property of the normal distribution, not of large samples. A simulation shows both effects: a straight line with  $n$  equally spaced design points on  $[0, 1]$ , prediction at  $x_0 = 0.8$ , and errors with mean zero and variance one that are normal, centred exponential (skewed) or uniform (short-tailed). With 20000 data sets for each case, the coverages of the nominal 80% and 95% intervals are:

| Errors      | $n$ | mean, 80% | mean, 95% | new obs.,<br>80% | new obs.,<br>95% |
|-------------|-----|-----------|-----------|------------------|------------------|
| normal      | 20  | 0.799     | 0.949     | 0.797            | 0.949            |
| normal      | 400 | 0.805     | 0.953     | 0.802            | 0.953            |
| exponential | 20  | 0.774     | 0.931     | 0.863            | 0.939            |
| exponential | 400 | 0.799     | 0.949     | 0.896            | 0.949            |
| uniform     | 20  | 0.797     | 0.948     | 0.787            | 0.983            |
| uniform     | 400 | 0.800     | 0.950     | 0.741            | 1.000            |

At  $n = 400$  the intervals for the mean are near nominal for every law, while the prediction intervals approach the limits of [Proposition 12.3.3](#): 0.898 and 0.948 (exponential), 0.740 and 1.000 (uniform). The exponential 95% interval looks fine only by accident: at  $n = 400$  it misses above in a proportion 0.051 of the samples and below in 0.0000, since a centred exponential error is never below  $-1$ .

```
rng = np.random.default_rng(1203)
laws = {
```

```

"normal": lambda size: rng.normal(size=size),
"exponential": lambda size: rng.exponential(size=size) - 1.0,      #
    skewed
"uniform": lambda size: rng.uniform(-np.sqrt(3), np.sqrt(3), size=size),  #
    short tails
}
    all: mean 0, variance 1
x0_sim = 0.8

def simulate(law, n_sim, reps=20_000):
    x = np.linspace(0.0, 1.0, n_sim)
    Xs = np.column_stack([np.ones(n_sim), x])
    Hs = np.linalg.solve(Xs.T @ Xs, Xs.T)          # beta_hat = Hs y
    z0 = np.array([1.0, x0_sim])
    h0 = z0 @ np.linalg.solve(Xs.T @ Xs, z0)
    E = laws[law](reps, n_sim)                    # true beta = 0,
    sigma = 1
    fit = E @ (Hs.T @ z0)
    s_sim = np.sqrt(np.sum((E - (E @ Hs.T) @ Xs.T) ** 2, axis=1) / (n_sim - 2))
    new = laws[law](reps)                          # the new
    observation's error
    out = {}
    for level in (0.80, 0.95):
        t = stats.t.ppf(0.5 + level / 2, n_sim - 2)
        out[("mean", level)] = np.mean(np.abs(fit) <= t * s_sim * np.sqrt(h0))
        out[("new", level)] = np.mean(np.abs(new - fit) <= t * s_sim * np.sqrt(1 +
h0))
    return out

table = {(law, m): simulate(law, m) for law in laws for m in (20, 400)}
for (law, m), out in table.items():
    print(f"{law:12s} n = {m:3d}: " + ", ".join(f"{k[0]} {k[1]:.2f}: {v:.3f}" for k,
v in out.items()))

```

[Run this code online](#)

Remedies, transformation ([Chapter 22](#)) and resampling ([Chapter 23](#)), come later. Checking normality matters much more for prediction than for estimation.

*Remark* (Prediction is not intervention). A prediction interval describes the next observation from the *same process*. If the regressors are set by intervention, say a policy that changes a state's poverty rate, the observed association need not describe what happens ([Chapter 25](#)).

## 12.3.4 Exercises

### 12.3.4.1 A. Check your understanding

#### Exercise. A1

Show that the ratio of the length of the prediction interval to that of the confidence interval for the mean at  $\mathbf{x}_0$  is  $\sqrt{(1 + h_0)/h_0}$ . What is it in [Example \(Predicting a state's murder](#)

rate), and what happens to it as  $n \rightarrow \infty$  with the design growing regularly?

*Solution.* Both intervals use the multiplier  $t_{n-r,\alpha/2} s$ , with  $\sqrt{1+h_0}$  and  $\sqrt{h_0}$  respectively. At the centroid,  $h_0 = 0.0200$  and the ratio is  $\sqrt{51} = 7.14$ . As the design grows,  $h_0 \rightarrow 0$  and the ratio tends to infinity: the confidence interval shrinks to a point while the prediction interval does not.

### 12.3.4.2 B. Practice

#### Exercise. B1

Two new observations will be made, at  $\mathbf{x}_{01}$  and  $\mathbf{x}_{02}$ , with independent errors. Find an exact  $1 - \alpha$  prediction interval for their difference  $Y_{01} - Y_{02}$ . Show that when  $\mathbf{x}_{01} = \mathbf{x}_{02}$  it does not depend on the data except through  $s$ .

*Solution.*  $Y_{01} - Y_{02} - (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \hat{\boldsymbol{\beta}}$  is normal with mean zero and variance  $\sigma^2\{2 + (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \mathbf{G}(\mathbf{x}_{01} - \mathbf{x}_{02})\}$ , independent of  $s^2$  by the argument of [Theorem 12.3.1](#). The interval is  $(\mathbf{x}_{01} - \mathbf{x}_{02})^\top \hat{\boldsymbol{\beta}} \pm t_{n-r,\alpha/2} s \sqrt{2 + (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \mathbf{G}(\mathbf{x}_{01} - \mathbf{x}_{02})}$ . When the two points coincide it is  $\pm t_{n-r,\alpha/2} s \sqrt{2}$ : the difference of two new errors, for which the fitted model contributes only its estimate of  $\sigma$ .

#### Exercise. B2

In [Proposition 12.3.2\(c\)](#), let  $\mathbf{X}$  have full column rank and add a single column  $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$ . Show that the new leverage is

$$h_{0+} = h_0 + \frac{(z_0 - \mathbf{k}^\top \mathbf{x}_0)^2}{\|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2}, \quad \mathbf{k} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{z}.$$

So the leverage is unchanged iff  $z_0$  equals the value  $\mathbf{k}^\top \mathbf{x}_0$  predicted for it by the other regressors.

*Solution.* Use [Proposition 12.3.2\(a\)](#) for the enlarged model. Write  $\mathbf{x}_0 = \mathbf{X}^\top \boldsymbol{\rho}$  and split any feasible  $\mathbf{a}$  as  $\mathbf{a} = \mathbf{M}\mathbf{a} + \mathbf{b}$  with  $\mathbf{b} \perp \mathcal{C}(\mathbf{X})$ . The constraint  $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$  forces  $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ , as in the proof of the proposition. The constraint  $\mathbf{z}^\top \mathbf{a} = z_0$  then reads  $\mathbf{z}^\top \mathbf{M}\boldsymbol{\rho} + \mathbf{z}^\top \mathbf{b} = z_0$ , and  $\mathbf{z}^\top \mathbf{M}\boldsymbol{\rho} = \mathbf{z}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\rho} = \mathbf{k}^\top \mathbf{x}_0$ . Since  $\mathbf{b} \perp \mathcal{C}(\mathbf{X})$ ,  $\mathbf{z}^\top \mathbf{b} = ((\mathbf{I} - \mathbf{M})\mathbf{z})^\top \mathbf{b}$ , and by the Cauchy–Schwarz inequality the shortest  $\mathbf{b}$  with  $((\mathbf{I} - \mathbf{M})\mathbf{z})^\top \mathbf{b} = z_0 - \mathbf{k}^\top \mathbf{x}_0$  is a multiple of  $(\mathbf{I} - \mathbf{M})\mathbf{z}$ , with  $\|\mathbf{b}\|^2 = (z_0 - \mathbf{k}^\top \mathbf{x}_0)^2 / \|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2$ . Adding  $\|\mathbf{M}\boldsymbol{\rho}\|^2 = h_0$  gives the formula.

#### Exercise. B3

For the line through the origin,  $EY_i = \beta x_i$  with  $\sum x_i^2 > 0$ , write the  $1 - \alpha$  confidence interval for the mean at  $x_0$  and the prediction interval for  $Y_0$ . Show that the confidence interval for the mean has length proportional to  $|x_0|$ , and is a point at  $x_0 = 0$ . Why is that reasonable,

and what does it assume?

*Solution.* Here  $r = 1$ ,  $\hat{\beta} = \sum x_i y_i / \sum x_i^2$ ,  $s^2 = \sum (y_i - \hat{\beta} x_i)^2 / (n - 1)$  and  $h_0 = x_0^2 / \sum x_i^2$ . The intervals are  $\hat{\beta} x_0 \pm t_{n-1, \alpha/2} s |x_0| / \sqrt{\sum x_i^2}$  and  $\hat{\beta} x_0 \pm t_{n-1, \alpha/2} s \sqrt{1 + x_0^2 / \sum x_i^2}$ . At  $x_0 = 0$  the model asserts that the mean is exactly zero, so there is nothing to estimate. The interval is only as good as that assertion: if the true line has a small intercept, the interval near  $x_0 = 0$  is too short.

#### Exercise. B4

Suppose the new regressor vector  $\mathbf{X}_0$  is random, independent of the pair  $(\mathbf{Y}, \varepsilon_0)$ , and always lies in  $\mathcal{C}(\mathbf{X}^\top)$ . Show that the prediction interval of [Theorem 12.3.1](#), computed at the realized  $\mathbf{X}_0$ , still covers  $Y_0$  with probability exactly  $1 - \alpha$ .

*Solution.* Condition on  $\mathbf{X}_0 = \mathbf{x}_0$ . By independence, given this event  $(\mathbf{Y}, \varepsilon_0)$  has its unconditional distribution, so [Theorem 12.3.1](#) applies and the conditional coverage is  $1 - \alpha$  for every  $\mathbf{x}_0$ . Averaging over the distribution of  $\mathbf{X}_0$  gives  $1 - \alpha$ .

#### 12.3.4.3 C. Going deeper

#### Exercise. C1

Let  $k$  new observations be made at the rows of a  $k \times p$  matrix  $\mathbf{X}_0$  whose rows lie in  $\mathcal{C}(\mathbf{X}^\top)$ , with errors independent of each other and of  $\mathbf{Y}$ . Show that  $\mathbf{D} = \mathbf{Y}_0 - \mathbf{X}_0 \hat{\beta} \sim N_k(\mathbf{0}, \sigma^2(\mathbf{I}_k + \mathbf{X}_0 \mathbf{G} \mathbf{X}_0^\top))$ , independent of  $s^2$ , and deduce that the intervals

$$\mathbf{x}_{0j}^\top \hat{\beta} \pm \sqrt{k F_\alpha(k, n - r)} s \sqrt{1 + h_{0j}}, \quad j = 1, \dots, k,$$

contain all  $k$  new observations simultaneously with probability at least  $1 - \alpha$ .

*Solution.* Normality, the mean and the covariance follow as in [Theorem 12.3.1](#), and  $\mathbf{S} = \mathbf{I}_k + \mathbf{X}_0 \mathbf{G} \mathbf{X}_0^\top$  is positive definite. So  $\mathbf{D}^\top \mathbf{S}^{-1} \mathbf{D} / (k s^2) \sim F(k, n - r)$  (Corollary 4.3.3 and Definition 4.1.2). By Lemma 12.2.3 with  $\mathbf{a} = \mathbf{e}_j$ , the event  $\mathbf{D}^\top \mathbf{S}^{-1} \mathbf{D} \leq k s^2 F_\alpha$  implies  $|D_j| \leq \sqrt{k F_\alpha} s \sqrt{S_{jj}}$  for every  $j$ , and  $S_{jj} = 1 + h_{0j}$ . [Chapter 13](#) compares this with Bonferroni's method, which is shorter when  $k$  is small.

#### Exercise. C2

A  $(P, 1 - \alpha)$  **tolerance interval** at  $\mathbf{x}_0$  is a random interval that, with probability at least  $1 - \alpha$  over  $\mathbf{Y}$ , contains at least a proportion  $P$  of the distribution of  $Y_0$ . Show that the prediction interval of [Theorem 12.3.1](#) is not a  $(1 - \alpha, 1 - \alpha)$  tolerance interval, by showing that its conditional coverage given  $\mathbf{Y}$  is below  $1 - \alpha$  with probability bounded away from zero. Construct a conservative tolerance interval from a lower confidence bound for the conditional

coverage, using a  $1 - \alpha/2$  upper confidence bound for  $\sigma$  and a  $1 - \alpha/2$  confidence interval for  $\mathbf{x}_0^\top \boldsymbol{\beta}$ .